

Sales Dashboard – Interview Explanation

Project Overview

This project is an end-to-end Sales Analytics dashboard developed using Power BI. The main objective of the project is to analyze sales performance across different dimensions such as time, region, category, sub-category, and customer segments. The dashboard helps identify growth trends, seasonal patterns, and loss-making areas to support business decision-making.

Data and Tools Used

The dataset consists of sales transaction data from 2014 to 2017. It includes attributes such as region, state, city, category, sub-category, segment, order date, sales, profit, and quantity. Power BI Desktop was used for data modeling, DAX calculations, and interactive visualizations.

Dashboard Structure

1. Overall Sales Performance

This page provides a high-level view of business performance. It includes KPIs such as Total Revenue of \$2.3M, Total Profit of \$286K, and Total Quantity Sold of 38K units. The analysis shows consistent year-on-year sales growth. However, profit growth is slower, indicating margin pressure. Quarterly analysis reveals strong seasonality, with Q4 generating the highest sales and profit each year.

2. Region and State-Level Analysis

This page focuses on geographical performance. It highlights that some cities such as Philadelphia and Houston generate high sales volumes but negative profit, indicating issues related to pricing, discounts, or operational costs. Technology performs well across most states, while the Furniture category shows weak profitability.

3. Category and Sub-Category Analysis

This page analyzes product-level performance. Technology emerges as the most profitable category, while Furniture contributes low or negative profit despite reasonable sales volumes. Sub-category analysis shows that not all high-selling products are profitable, helping identify areas for optimization.

Key Business Insights

1. Revenue is growing, but profit margins require improvement.
2. Technology is the strongest and most profitable category.
3. Furniture category needs pricing or cost optimization.
4. The Consumer segment contributes more than 50% of total revenue.
5. The business shows strong Q4 seasonality.
6. Certain cities are loss-making despite high sales volume.

Business Impact

This dashboard helps management focus on profitable categories, identify loss-making regions, optimize pricing strategies, plan inventory based on seasonal demand, and improve overall profitability.